

Lalith Kumar Dabilpuram

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SUMMARY

Senior AI / ML Engineer with 5+ years of experience building predictive ML models, production RAG pipelines, and multi-agent systems using Python, Scikit-learn, XGBoost, LangChain, LangGraph, and CrewAI. Proven track record improving model accuracy, automating document workflows, and deploying ML/LLM solutions on GCP with MLflow and FastAPI. Skilled in feature engineering, agentic orchestration, and end-to-end MLOps for enterprise-scale AI and ML solutions.

TECHNICAL SKILLS

Programming: Python, SQL, R, Java, TypeScript, PySpark, Pandas, NumPy
Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, Random Forest, Feature Engineering
Deep Learning & NLP: CNNs, Transformers, Transfer Learning, spaCy, NLTK, BERT, Text Classification
Generative AI & LLMs: OpenAI, Claude, Hugging Face, Prompt Engineering, RAG Pipelines, Agentic AI
AI Frameworks & Orchestration: LangChain, LangGraph, CrewAI, LlamaIndex, Fine-Tuning, Ragas
Vector Databases & Search: FAISS, ChromaDB, Pinecone, Embeddings, Semantic Search, Metadata Filtering
Computer Vision & OCR: OpenCV, YOLO, OCR, Tesseract, AWS Textract, Object Detection, Image Segmentation
Data Engineering & MLOps: Apache Spark, dbt, Databricks, ETL/ELT Pipelines, MLflow, Model Monitoring
Cloud, APIs & Infrastructure: GCP (Vertex AI, Cloud Run), AWS, FastAPI, Docker, CI/CD, Kubernetes
Databases, Visualization & Tools: PostgreSQL, MySQL, Tableau, Power BI, Git, Jupyter, Agile, JIRA

EXPERIENCE

AI / ML Engineer

Nov 2023 – Present

Deloitte

VA, USA

- Developed RAG pipelines using LangChain, OpenAI embeddings, and ChromaDB with agentic retrieval routing, improving response accuracy by 28% and reducing manual research time through agent orchestration.
- Built multi-agent workflows using CrewAI and LangGraph integrating RAG for document analysis and task routing, reducing analyst effort by 22% via automated retrieval-augmented reasoning and monitoring.
- Designed agentic workflows with FastMCP and FastAPI microservices for LLM inference, vector search, and multi-agent orchestration, integrating LangSmith for evaluation, observability, monitoring in production.
- Built predictive ML models using Python, Scikit-learn, and XGBoost for enterprise analytics, applying cross-validation and hyperparameter tuning to select top algorithms, improving forecast accuracy by 18%.
- Implemented OCR pipelines using Document AI and Tesseract integrated with agentic extraction workflows, reducing manual data entry by 42% through intelligent document processing and automated categorization.
- Deployed ML and LLM models to production on Google Cloud using Vertex AI and Cloud Run with Docker, MLflow tracking and versioning to ensure reliability and reproducibility across enterprise workflows.
- Built model monitoring and evaluation pipelines tracking prediction drift, latency, and quality metrics, enabling automated retraining and sustained model performance and reliability in production deployments.

Data Scientist

March 2020 – July 2023

Broadridge Financial Solutions

Hyderabad, India

- Built a classification model using Python, Scikit-learn, and XGBoost to flag anomalous transactions, applying cross-validation and feature engineering to improve detection accuracy by 20% across operations.
- Developed a regression model using Python and Scikit-learn to forecast transaction volumes, applying feature engineering and cross-validation to support capacity planning and improve forecast accuracy by 21%.
- Performed exploratory data analysis and statistical testing on financial datasets using Python, Pandas, SQL, identifying significant features and correlations that improved model design and prediction accuracy.
- Built end-to-end ML pipelines using Python, PySpark, and Scikit-learn for training and validating predictive models on financial datasets, improving processing efficiency by 24% across reporting workflows.
- Validated ML models using statistical techniques including hypothesis testing and significance analysis, documenting assumptions and metrics to support model governance and audit compliance standards.

EDUCATION

Master of Science in Computer Science

Aug 2023 – May 2025

University of Central Missouri

MO, USA